NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
THEORY EXAMINATION

Question paper (**Evening session: CS, IT & ECE**)

Month and Year of the Examination: **Jan, 2023**

Programme: **B. Tech (NST)** Semester...**7th**

Subject: **Nanoscience and Technology (NST)**

Course No. **PHOE15** Maximum Marks.....**50**

No. of Questions to be Attempted.....**05** Time allowed.....**3 Hrs**

Total No. of Questions.....**05** Total No of pages used...**01**

Unless stated otherwise, the symbols have their usual meanings in context with the Subject. Assume suitably and state, additional data required, if any.

The Candidates, before starting to write the solutions, should please check the Question Paper for any discrepancy, and also ensure that they have been delivered the question paper of **right course no.** and right subject title.

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1. (i) Explain why the filter is required in XRD power diffraction method. **[2]**

(ii) State the differences between SEM and TEM. **[2]**

(iii) What is the selection rule for a reflection to present for a BCC structure? **[2]**

(iv) Give examples of one, two and three dimensional nanomaterials. **[2]**

(v) What is Quantum confinement? **[2]**

2. (a) What is bottom-up approach in the synthesis of nanomaterials? Give its advantages and disadvantages. **[7]**

(b) Explain why band gap of nanomaterials increases with size reduction.? **[3]**

3. Write in brief on e-beam evaporation. Distinguish between thermal evaporation and e-beam evaporation. **[10]**

4. (a) Explain why vacuum is basic requirement for an electron microscope operation? What are the most significant advantages of SEM in comparison to optical microscope? [2.5+2.5]

(b) What are secondary and backscattered electrons, and what is their origin? **[5]**

OR

What is the importance of crystal structure determination in materials? Discuss the various methods in detail for the structure determination using X-ray diffraction. **[10]**

5. Discuss the applications of nanomaterials in the following fields: **[10]**

(i) Data storage (ii) Photocatalytic (ii) Nanoelectronics (iv) Solar energy conversion

OR

Explain in detail the principle, working and application of Raman Spectroscopy for the evaluation of properties of nano materials and nano structures. **[10]**

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